

1. Generating and Characteristic Functions

Probability generating function (pgf)

For X nonnegative integer-valued:

$$G_X(s) \equiv \mathbb{E}(s^X) = \sum_{k \geq 0} s^k \mathbb{P}(X = k).$$

Useful facts:

$$G_X(0) = \mathbb{P}(X = 0), \quad G_X(1) = 1,$$

$$\mathbb{P}(X = k) = \frac{G_X^{(k)}(0)}{k!},$$

$$\mathbb{E}(X) = G'_X(1),$$

$$\mathbb{E}(X(X-1) \cdots (X-r+1)) = G_X^{(r)}(1).$$

If $X_1, \dots, X_n$ are independent nonnegative integer-valued and

$$Y = X_1 + \cdots + X_n,$$

then

$$G_Y(s) = \prod_{j=1}^n G_{X_j}(s).$$

If additionally X_j are equally distributed:

$$G_Y(s) = (G_X(s))^n.$$

Moment generating function (mgf)

$$\Phi_X(t) \equiv \mathbb{E}(e^{tX}).$$

If X is discrete:

$$\Phi_X(t) = \sum_i e^{tx_i} \mathbb{P}(X = x_i).$$

If X is continuous:

$$\Phi_X(t) = \int_{-\infty}^{\infty} e^{tx} f_X(x) dx.$$

If X is nonnegative integer-valued:

$$\Phi_X(t) = G_X(e^t).$$

Convolution: if $S = X_1 + \cdots + X_n$ with independent X_j ,

$$\Phi_S(t) = \prod_{j=1}^n \Phi_{X_j}(t).$$

Series expansion around 0:

$$\Phi_X(t) = \sum_{k=0}^{\infty} \frac{\mathbb{E}(X^k)}{k!} t^k$$

whenever the moments exist and the series is valid. Hence

$$\Phi_X^{(k)}(0) = \mathbb{E}(X^k).$$

Uniqueness: if $\Phi_X(t) = \Phi_Y(t)$ for $|t| < h$ for some $h > 0$, then X, Y are identically distributed.

Characteristic function (cf)

$$M_X(\omega) \equiv \mathbb{E}(e^{i\omega X}).$$

If X is discrete:

$$M_X(\omega) = \sum_k e^{i\omega x_k} \mathbb{P}(X = x_k).$$

If X is continuous:

$$M_X(\omega) = \int_{-\infty}^{\infty} e^{i\omega x} f_X(x) dx.$$

Basic properties:

$$M_X(0) = 1, \quad |M_X(\omega)| \leq 1, \quad M_X(-\omega) = \overline{M_X(\omega)}.$$

Affine transformation: if $Y = aX + b$,

$$M_Y(\omega) = e^{i\omega b} M_X(a\omega).$$

Convolution: if $Y = X_1 + \cdots + X_n$ with independent X_j ,

$$M_Y(\omega) = \prod_{j=1}^n M_{X_j}(\omega).$$

Moments: if all needed moments exist,

$$M_X(\omega) = \sum_{k=0}^{\infty} \frac{i^k \mathbb{E}(X^k)}{k!} \omega^k, \quad M_X^{(k)}(0) = i^k \mathbb{E}(X^k).$$

In particular

$$\mathbb{E}(X) = \frac{M'_X(0)}{i}, \quad \mathbb{E}(X^2) = -M''_X(0), \quad \text{Var}(X) = -M''_X(0) - \left(\frac{M'_X(0)}{i} \right)^2$$

Inversion and uniqueness

If X is continuous with density f_X ,

$$f_X(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega x} M_X(\omega) d\omega$$

at points where f_X is differentiable.

If X is integer-valued,

$$\mathbb{P}(X = k) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-ik\omega} M_X(\omega) d\omega.$$

Hence the characteristic function uniquely determines the law.

Joint characteristic function

For a random vector $X = (X_1, \dots, X_n)^t$,

$$M_X(\omega_1, \dots, \omega_n) = \mathbb{E}\left(e^{i(\omega_1 X_1 + \cdots + \omega_n X_n)}\right).$$

If $X_1, \dots, X_n$ are independent,

$$M_X(\omega_1, \dots, \omega_n) = \prod_{j=1}^n M_{X_j}(\omega_j).$$

2. Branching Processes

Galton–Watson process

$$Z_0 = 1, \quad Z_{n+1} = X_1 + \cdots + X_{Z_n},$$

where the offspring variables X_i are i.i.d. with

$$\mathbb{P}(X = i) = p_i, \quad i \geq 0.$$

Let

$$G_X(s) = \sum_{i \geq 0} p_i s^i.$$

Conditional law:

$$\mathbb{P}(Z_{n+1} = m \mid Z_n = r) = \mathbb{P}(X_1 + \cdots + X_r = m).$$

pgf of Z_n

Let

$$G_n(s) \equiv \mathbb{E}(s^{Z_n}).$$

Then

$$G_{n+1}(s) = G_n(G_X(s)).$$

Hence

$$G_1(s) = G_X(s), \quad G_n(s) = G_X^{(n)}(s)$$

(n -fold composition).

Extinction

Define

$$d_n = \mathbb{P}(Z_n = 0).$$

Then

$$d_n = G_X(d_{n-1}), \quad d_0 = 0.$$

The probability of ultimate extinction is

$$d = \lim_{n \rightarrow \infty} d_n.$$

It satisfies

$$d = G_X(d),$$

and d is the *smallest nonnegative root* of $s = G_X(s)$.

If

$$m = \mathbb{E}(X) = G'_X(1),$$

then:

$$m \leq 1 \implies d = 1, \quad m > 1 \implies d < 1.$$

So survival probability is

$$\mathbb{P}(\text{non-extinction}) = 1 - d.$$

Expected generation size

Let

$$m_n = \mathbb{E}(Z_n), \quad m = \mathbb{E}(X).$$

Since $m_n = G'_n(1)$, from $G_{n+1}(s) = G_n(G_X(s))$:

$$m_{n+1} = m_n m,$$

hence

$$\mathbb{E}(Z_n) = m_n = m^n.$$

Practical summary

- Mean offspring $m = G'_X(1)$.
- Extinction probability = smallest root of $s = G_X(s)$.
- If $m \leq 1$: extinction sure.
- If $m > 1$: positive survival probability.
- pgf of Z_n : $G_n(s) = G_X^{(n)}(s)$.
- Mean population at generation n : $\mathbb{E}(Z_n) = m^n$.

3. The Multivariate Gaussian Distribution

Covariance matrix

For $X = (X_1, \dots, X_n)^t$,

$$m_X = \mathbb{E}(X), \qquad K_X = \mathbb{E}((X - m_X)(X - m_X)^t).$$

Entries:

$$(K_X)_{ij} = \text{Cov}(X_i, X_j), \qquad (K_X)_{ii} = \text{Var}(X_i).$$

Properties:

$$K_X = K_X^t, \qquad z^t K_X z \geq 0 \quad \forall z \in \mathbb{R}^n.$$

So K_X is symmetric positive semidefinite.

Moreover,

$$z^t K_X z = \text{Var}(z^t X).$$

Gaussian random vector

$X \in \mathbb{R}^n$ is Gaussian if its characteristic function is

$$M_X(\omega) = \exp\left(i\omega^t m - \frac{1}{2}\omega^t K \omega\right), \qquad \omega \in \mathbb{R}^n,$$

with $m \in \mathbb{R}^n$ and K symmetric positive semidefinite. Notation:

$$X \sim \text{N}(m, K).$$

Marginals

If $X \sim \text{N}(m, K)$, then each component is Gaussian:

$$X_j \sim \text{N}(m_j, k_{jj}).$$

Also

$$\text{Cov}(X_r, X_s) = k_{rs}.$$

Linear transformations

If

$$Y = AX + b,$$

where A is $m \times n$ and $b \in \mathbb{R}^m$, then

$$Y \sim \text{N}(Am_X + b, AK_X A^t).$$

Characterization by linear combinations

A random vector X is Gaussian iff

$$a^t X$$

is a 1-dimensional Gaussian random variable for every $a \in \mathbb{R}^n$.

Uncorrelated Gaussian $\Rightarrow$ independent

If $X_1, \dots, X_n$ are jointly Gaussian and pairwise uncorrelated, then they are jointly independent.

Eigenvalues and invertibility

Since K_X is symmetric,

$$CK_X C^t = D = \text{diag}(\lambda_1, \dots, \lambda_n)$$

for some orthogonal C . Then

$$\lambda_i \geq 0.$$

For a Gaussian vector, the following are equivalent:

$$\begin{aligned} K_X \text{ positive definite} &\iff \lambda_i > 0 \quad \forall i \\ &\iff \det(K_X) > 0 \\ &\iff K_X^{-1} \text{ exists} \\ &\iff f_X \text{ exists.} \end{aligned}$$

Density formula

If $X \sim \text{N}(m, K)$ and $\det(K) > 0$, then

$$f_X(x) = \frac{1}{(2\pi)^{n/2} \sqrt{\det(K)}} \exp\left(-\frac{1}{2}(x - m)^t K^{-1}(x - m)\right).$$

For $n = 2$, with means m_X, m_Y , variances σ_X^2, σ_Y^2 , correlation ρ ,

$$f_{XY}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)}a(x, y)\right),$$

where

$$a(x, y) = \left(\frac{x - m_X}{\sigma_X}\right)^2 - 2\rho\left(\frac{x - m_X}{\sigma_X}\right)\left(\frac{y - m_Y}{\sigma_Y}\right) + \left(\frac{y - m_Y}{\sigma_Y}\right)^2.$$

Conditional law in the bivariate case

If (X, Y) is jointly Gaussian, then

$$Y \mid X = x \sim \text{N}(m_{Y|X}, \sigma_{Y|X}^2),$$

with

$$m_{Y|X} = m_Y + \rho \frac{\sigma_Y}{\sigma_X}(x - m_X), \qquad \sigma_{Y|X}^2 = (1 - \rho^2)\sigma_Y^2.$$

Quadratic form

If $X \sim \text{N}(m, K)$ and $\det(K) > 0$, then

$$(X - m)^t K^{-1}(X - m) \sim \chi^2(n).$$

4. Sequences of Random Variables

Modes of convergence

In probability

$$X_n \xrightarrow{P} X \iff \forall \varepsilon > 0, \quad \mathbb{P}(|X_n - X| \geq \varepsilon) \rightarrow 0.$$

In distribution

$$X_n \xrightarrow{d} X \iff F_{X_n}(x) \rightarrow F_X(x)$$

at every continuity point of F_X .

In mean square

$$X_n \xrightarrow{2} X \iff \mathbb{E}((X_n - X)^2) \rightarrow 0.$$

In r -mean

$$X_n \xrightarrow{r} X \iff \mathbb{E}(|X_n - X|^r) \rightarrow 0.$$

Almost surely

$$X_n \xrightarrow{\text{a.s.}} X \iff \mathbb{P}(\{\omega : X_n(\omega) \rightarrow X(\omega)\}) = 1.$$

Weak law of large numbers (WLLN)

If $X_1, X_2, \dots$ are i.i.d. with finite mean m and finite variance, and

$$\overline{X}_n = \frac{X_1 + \dots + X_n}{n},$$

then

$$\overline{X}_n \xrightarrow{P} m.$$

Chebyshev proof:

$$\mathbb{E}(\overline{X}_n) = m, \qquad \text{Var}(\overline{X}_n) = \frac{\text{Var}(X_1)}{n},$$

so

$$\mathbb{P}(|\overline{X}_n - m| \geq \varepsilon) \leq \frac{\text{Var}(X_1)}{n\varepsilon^2} \rightarrow 0.$$

Strong law of large numbers (SLLN)

If $X_1, X_2, \dots$ are i.i.d., then

$$\overline{X}_n \xrightarrow{\text{a.s.}} m$$

for some constant m iff

$$\mathbb{E}(|X_1|) < \infty.$$

In that case,

$$m = \mathbb{E}(X_1).$$

Central limit theorem (CLT)

If $X_1, X_2, \dots$ are i.i.d. with

$$\mathbb{E}(X_k) = m, \qquad \text{Var}(X_k) = \sigma^2,$$

define

$$Z_n = \frac{\sum_{k=1}^n X_k - nm}{\sqrt{n}\sigma} = \frac{\overline{X}_n - m}{\sigma/\sqrt{n}}.$$

Then

$$Z_n \xrightarrow{d} \text{N}(0, 1).$$

Equivalently,

$$\overline{X}_n \approx \text{N}\left(m, \frac{\sigma^2}{n}\right) \quad \text{for large } n.$$

Continuity theorem

Let $M_n(\omega)$ be the characteristic functions of X_n .

If

$$M_n(\omega) \rightarrow M(\omega)$$

for all ω , and M is continuous at 0, then M is a characteristic function and

$$X_n \xrightarrow{d} X$$

for the corresponding X .

Conversely, if $X_n \xrightarrow{d} X$, then

$$M_n(\omega) \rightarrow M_X(\omega) \quad \forall \omega.$$

De Moivre–Laplace

If

$$S_n \sim \text{Bin}(n, p),$$

then

$$\frac{S_n - np}{\sqrt{npq}} \xrightarrow{d} \text{N}(0, 1).$$

Poisson theorem

If
$$X_n \sim \text{Bin}(n, \lambda/n),$$

then
$$X_n \xrightarrow{d} \text{Po}(\lambda).$$

Relations between convergence concepts

$$\begin{aligned} X_n &\xrightarrow{\text{a.s.}} X \implies X_n \xrightarrow{P} X, \\ X_n &\xrightarrow{r} X \implies X_n \xrightarrow{P} X \quad (r \geq 1), \\ X_n &\xrightarrow{P} X \implies X_n \xrightarrow{d} X, \\ r > s \geq 1, \quad X_n &\xrightarrow{r} X \implies X_n \xrightarrow{s} X. \end{aligned}$$

Useful converse facts:

$$X_n \xrightarrow{d} c \implies X_n \xrightarrow{P} c \quad \text{if } c \text{ is constant.}$$

If $X_n \xrightarrow{P} X$ and $|X_n| \leq C$ a.s. for all n , then

$$X_n \xrightarrow{r} X \quad \forall r \geq 1.$$

If for every $\varepsilon > 0$,

$$\sum_{n=1}^{\infty} \mathbb{P}(|X_n - X| > \varepsilon) < \infty,$$

then
$$X_n \xrightarrow{\text{a.s.}} X.$$

Operations with limits

If
$$X_n \xrightarrow{P} a, \quad Y_n \xrightarrow{P} b,$$

then
$$X_n + Y_n \xrightarrow{P} a + b, \quad X_n Y_n \xrightarrow{P} ab.$$

If $b \neq 0$,

$$\frac{X_n}{Y_n} \xrightarrow{P} \frac{a}{b}.$$

If g is continuous at a and $X_n \xrightarrow{P} a$, then
$$g(X_n) \xrightarrow{P} g(a).$$

Likewise, if $X_n \xrightarrow{d} X$ and g is continuous, then
$$g(X_n) \xrightarrow{d} g(X).$$

Delta method

If
$$a_n(X_n - \theta) \xrightarrow{d} N(0, \sigma),$$
 with $a_n \rightarrow \infty$, and g is differentiable at θ with $g'(\theta) \neq 0$, then

$$a_n(g(X_n) - g(\theta)) \xrightarrow{d} N(0, |g'(\theta)|^2 \sigma^2).$$

Borel–Cantelli lemmas

For events $A_1, A_2, \dots$, define

$$A^\star = \limsup_n A_n = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k.$$

Interpretation:

$$\omega \in A^\star \iff \omega \text{ belongs to infinitely many } A_n.$$

Then:

$$\begin{aligned} \sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty &\implies \mathbb{P}(A^\star) = 0, \\ \sum_{n=1}^{\infty} \mathbb{P}(A_n) = \infty \text{ and } A_n \text{ independent} &\implies \mathbb{P}(A^\star) = 1. \end{aligned}$$

So for independent events:

$$\mathbb{P}(A_n \text{ i.o.}) = \begin{cases} 0, & \sum_n \mathbb{P}(A_n) < \infty, \\ 1, & \sum_n \mathbb{P}(A_n) = \infty. \end{cases}$$

Very short exam checklist

- WLLN: finite mean + variance $\Rightarrow \overline{X}_n \xrightarrow{P} m$.
- SLLN: $\mathbb{E}|X_1| < \infty \iff \overline{X}_n \xrightarrow{\text{a.s.}} \mathbb{E}(X_1)$.
- CLT: standardized sum/sample mean $\Rightarrow N(0, 1)$.
- a.s. $\Rightarrow P \Rightarrow d$, and $L^r \Rightarrow P$.
- Borel–Cantelli decides whether events happen infinitely often.

Common Distributions (Summary)

Law	pmf/pdf	$G_X(s)$	$\Phi_X(t)$	$M_X(\omega)$
Be(p)	$\mathbb{P}(X = 0) = q, \mathbb{P}(X = 1) = p$	$q + ps$	$q + pe^t$	$q + pe^{i\omega}$
Bin(n, p)	$\mathbb{P}(X = k) = \binom{n}{k} p^k q^{n-k}$	$(q + ps)^n$	$(q + pe^t)^n$	$(q + pe^{i\omega})^n$
Po(λ)	$\mathbb{P}(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}$	$e^{\lambda(s-1)}$	$e^{\lambda(e^t-1)}$	$e^{\lambda(e^{i\omega}-1)}$
Ge(p)	$\mathbb{P}(X = k) = q^{k-1} p$	$\frac{ps}{1-qs}$	$\frac{pe^t}{1-qe^t}$	$\frac{pe^{i\omega}}{1-qe^{i\omega}}$
NBin(k, p)	$\mathbb{P}(X = n) = \binom{n-1}{k-1} p^k q^{n-k}$	$\left(\frac{ps}{1-qs}\right)^k$	$\left(\frac{pe^t}{1-qe^t}\right)^k$	$\left(\frac{pe^{i\omega}}{1-qe^{i\omega}}\right)^k$
Exp(μ)	$f_X(x) = \mu e^{-\mu x}, \ x \geq 0$	—	$\frac{\mu}{\mu - t}, \ t < \mu$	$\frac{\mu}{\mu - i\omega}$
N(0, 1)	$\frac{1}{\sqrt{2\pi}} e^{-x^2/2}$	—	$e^{t^2/2}$	$e^{-\omega^2/2}$
N(m, σ^2)	$\frac{1}{\sqrt{2\pi}\sigma} e^{-(x-m)^2/(2\sigma^2)}$	—	$\exp\left(mt + \frac{\sigma^2 t^2}{2}\right)$	$\exp\left(i\omega m - \frac{\sigma^2 \omega^2}{2}\right)$